How can we ensure the model is fair?

Responsible AI: Human-Centered Design

START → DESIGN → **DEVELOPMENT** → DEPLOYMENT → END

Course 1
Fundamentals of TinyML

- **What** am I building?
- **Who** am I building this for?
- What are the **consequences** for the user if it ***fails***?

Course 2
Applications of TinyML

- What data will be collected to train the model?
- Is the dataset biased?
- **How can we ensure the model is fair?**

Course 3
Deploying TinyML

Unfairness in ML

Model exhibits **discriminatory biases**, perpetuates **inequality** or performs less well for historically **disadvantaged groups**



- ***All ML discriminates*** (it just means to recognize a distinction, differentiate)
- Fairness is concerned with **wrongful** discrimination


Age
Disability
Medical History
Marital Status
Protected Classes
Sex
Race
Religion
National Origin

Discrimination

Disparate **Treatment**:

Membership in a protected class is used as an input to the model, decisions are differentiated on that basis in a way that disadvantages members of a protected class

Disparate **Impact**:

Outcomes of the model disproportionately disadvantage members of a protected class

1. Group Unawareness

Sensitive attributes are **not** included as features of the data (e.g. race, gender)

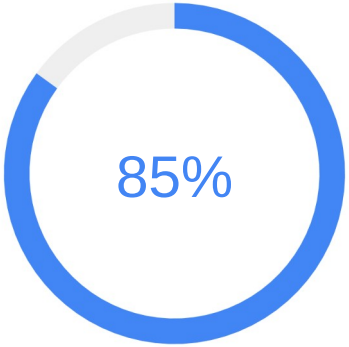


Pro: Avoids disparate treatment

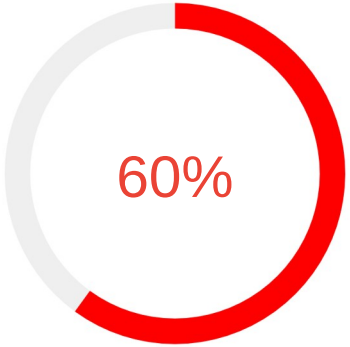
Con: Possibility of highly correlated features that are proxies of the sensitive attribute

2. Group Threshold

Counteract historical biases in data by **adjusting** confidence thresholds ***independently*** for each group



Group A



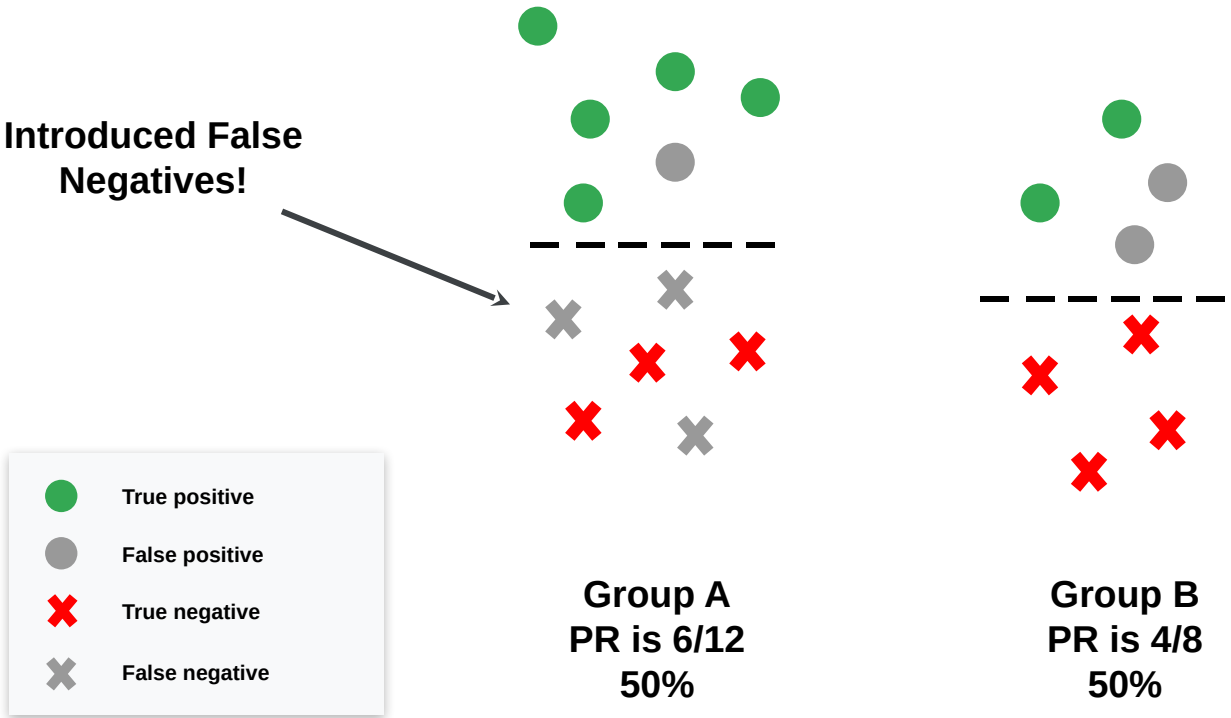
Group B

3. Demographic Parity

	Actually Healthy = Yes	Actually Healthy = No
Predicted Healthy = Yes	***True Positive***	***False Positive***
Predicted Healthy = No	False Negative	True Negative

The positive rate is the same across groups

Problem with Demographic Parity

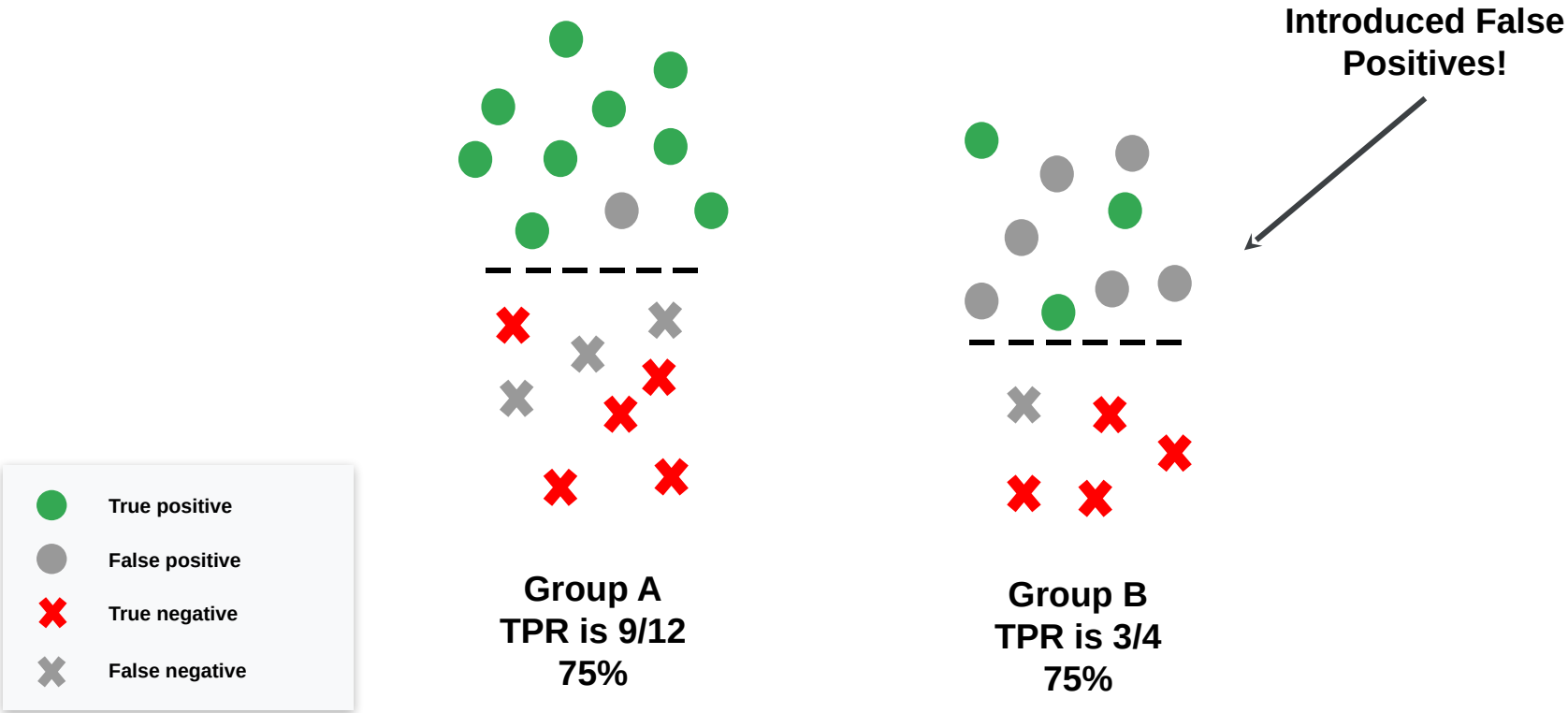


4. Equal Opportunity

	Actually Healthy = Yes	Actually Healthy = No
Predicted Healthy = Yes	***True Positive***	False Positive
Predicted Healthy = No	***False Negative***	True Negative

Qualified individuals should have an equal chance of being correctly classified for a desirable outcome.

Problem with Equality of Opportunity

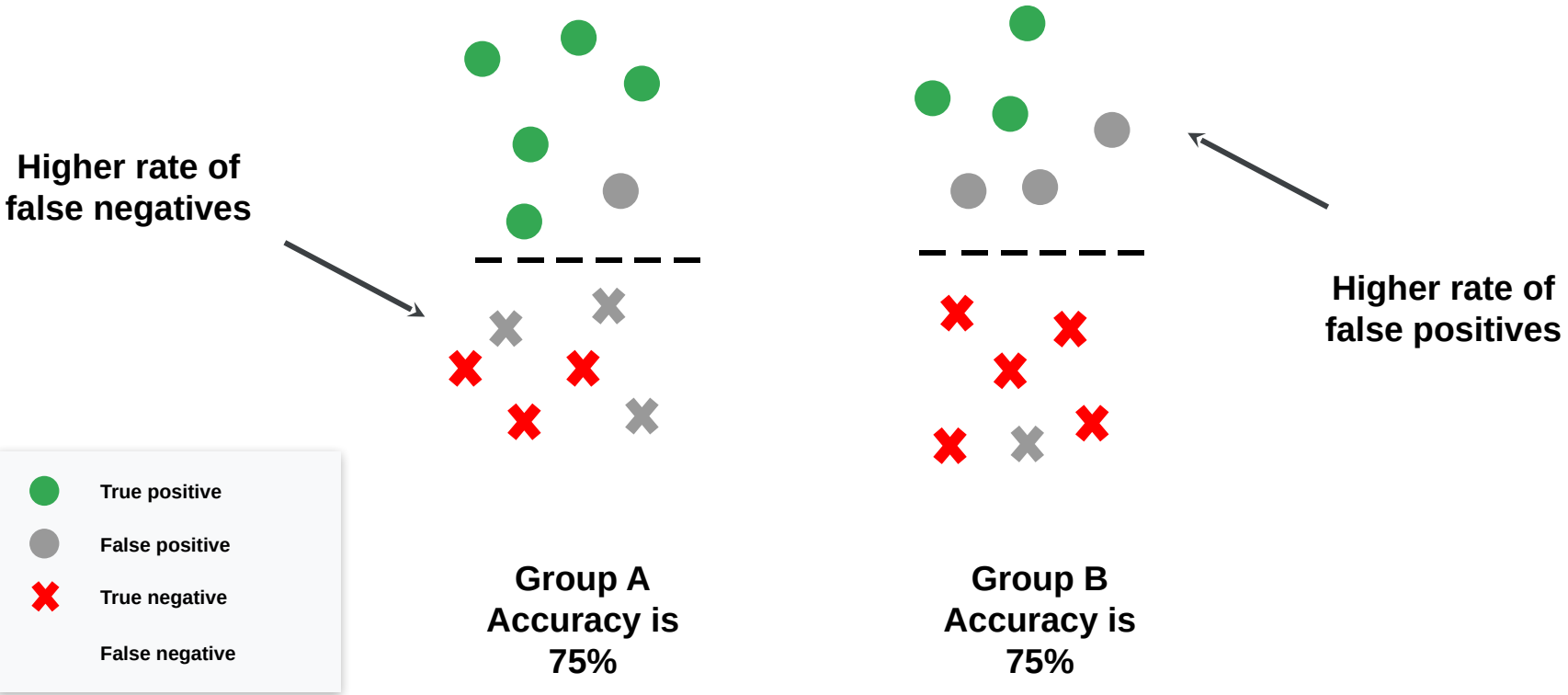


4. Equal Accuracy

	Actually Disease = Yes	Actually Disease = No
Predicted Disease = Yes	***True Positive***	False Positive
Predicted Disease = No	False Negative	***True Negative***

The percentage of correct classifications should be the same for all individuals

Problem with Equal Accuracy



Impossibility Theorem

We cannot satisfy all fairness metrics

at the same time!



For example:

- **Group Unawareness** is incompatible with **Group Threshold**
- **Equal Opportunity** is incompatible with **Equal Accuracy**

How can we mitigate ***unfairness*** in ML?

The Framing Trap

Algorithmic Frame

Do properties of the output match the input? Does the algorithm provide good accuracy on unseen data?

Data Frame

Has bias been removed from the training data? Does the demographic information of the data require optimization of the model?

Sociotechnical Frame

How does the model operate when considered as part of a system of humans and social institutions?

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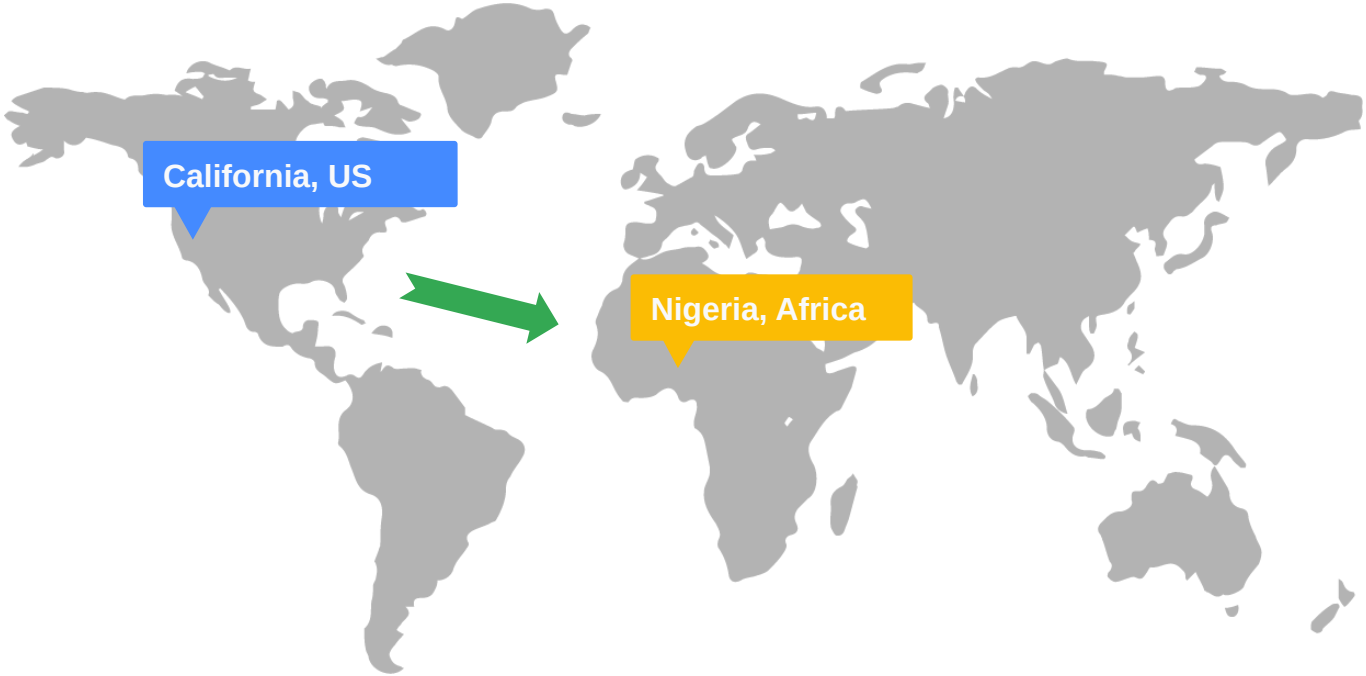
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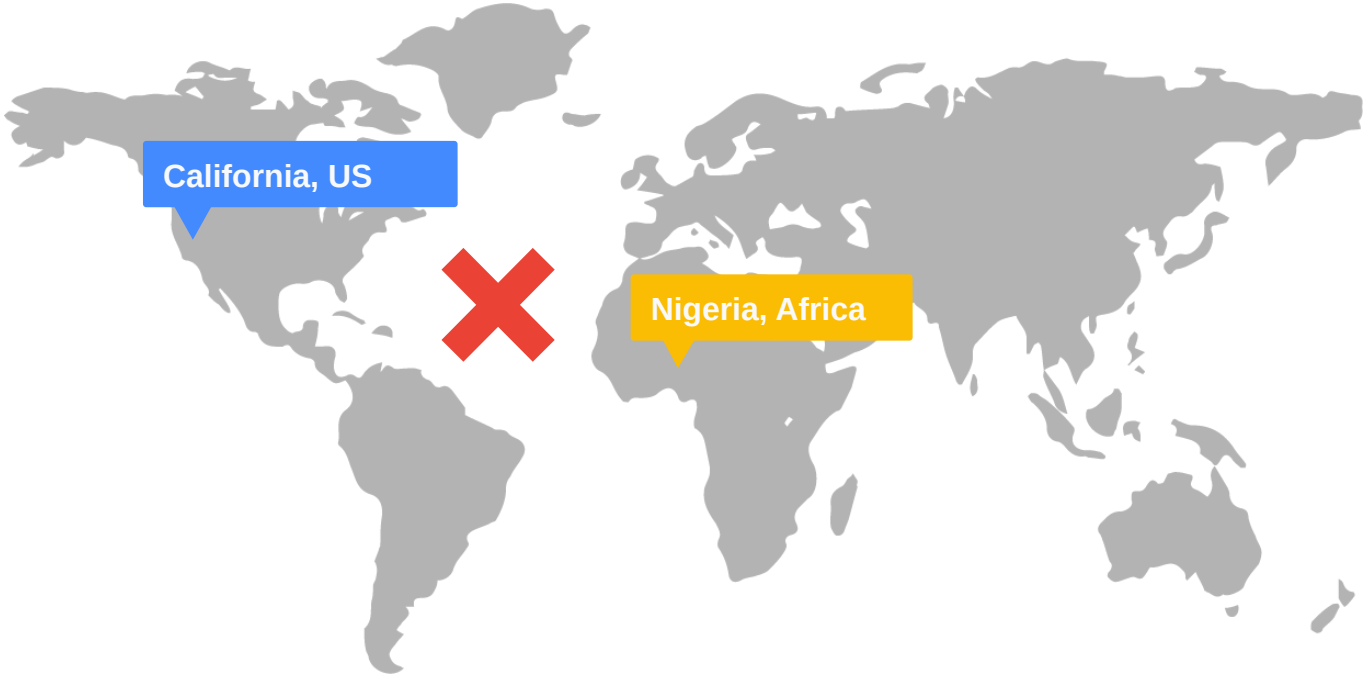
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The Portability Trap



The Portability Trap



Context Matters!

Repurposing algorithmic solutions may not preserve fair outcomes.

The **Formalism** Trap

Which **mathematical definition** of fairness should I choose?



Google’s **What-If Tool**

